ATISH GHOSH, Ph.D.

Computational & Theoretical Chemist | DFT - AIMD - NAMD - Modeling of Functional Nanomaterials

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PROFESSIONAL SUMMARY

Computational and theoretical chemist with a Ph.D. in Chemistry, specializing in Density Functional Theory (DFT), ab initio molecular dynamics (AIMD), and non-adiabatic molecular dynamics (NAMD) applied to low-dimensional and nanostructured materials. Research interests span photovoltaics, photocatalysis, electrocatalysis, spintronics, and metal-ion batteries, with 27 publications to date. Currently a postdoctoral researcher at the University of Massachusetts Amherst, working on machine-learning-accelerated atomistic simulation and electrochemical interface modeling.

RESEARCH INTERESTS

- Density Functional Theory (DFT) and ab initio modeling of low-dimensional and nanostructured materials
- Non-adiabatic molecular dynamics (NAMD) and ab initio molecular dynamics (AIMD) of photoexcited carrier relaxation
- Machine-learning and graph-neural-network-accelerated atomistic simulation
- 2D materials, MOFs/COFs, and van der Waals heterostructures for photocatalysis and electrocatalysis
- Metal-ion (Li/Na) battery electrode materials
- Spintronics and magnetic nanomaterials
- Halide perovskite and quantum-dot optoelectronic materials

EDUCATION

Ph.D. in Chemistry — Visva-Bharati (A Central University), India *August 2019 – August 2024*

Thesis: "Electronic Structure and Transport Properties of Low Dimensional Systems with Special Emphasis on Spintronics and Energy Applications." Supervisor: Prof. Dr. Pranab Sarkar.

M.Sc. in Chemistry (Physical Chemistry) — University of Burdwan, India *2016 – 2018*

B.Sc. in Chemistry — University of Burdwan, India *2013 – 2016*

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow — Department of Chemical & Biomolecular Engineering, University of Massachusetts Amherst, USA

March 2026 – Present

Working with Prof. Alexandra Zagalskaya on atomistic and electronic-structure modeling of electrochemical interfaces and electrocatalytic systems, bridging theory and experiment for the rational design of next-generation electrocatalysts.

Postdoctoral Research Associate — Department of Chemistry, Wayne State University, Detroit, MI, USA

April 2025 – March 2026

Worked in Prof. Zhenfei Liu's group developing electronic structure theory of cadmium chalcogenide quantum dot gels and two-dimensional perovskites, including energy-transfer dynamics at quantum dot–dye hybrid interfaces.

Doctoral Researcher — Department of Chemistry, Visva-Bharati (A Central University), India

August 2019 – August 2024

Conducted DFT- and NAMD-based research under Prof. Pranab Sarkar on the electronic structure, charge/energy transfer, and transport properties of low-dimensional systems for spintronic and energy applications, resulting in 17 peer-reviewed publications.

PUBLICATIONS (27 PEER-REVIEWED ARTICLES)

1. Understanding the impact of $BiVO_4$ surface facet and termination on infrared spectroscopy and oxygen evolution reaction, S Zhang, **A Ghosh**, H Maekawa, A Zagalskaya, R Donapati, S Ardo, Y Yamaguchi, A Kudo, N Ge, T A Pham, T Ogitsu, **In press, JCP**.
2. C. Tang, D. Ahire, A. Zagalskaya, **A. Ghosh**, A. Wang, Y. Hang, X. Chen, et al. "Stabilization of Oxygen Vacancies and Enhancement of Photocatalytic Nitrogen Fixation in Antimony Molybdate via Cobalt/Iron Co-Doping." Chemical Engineering Journal, 180400 (2026).
3. B. Ball, P. Roy, P. Das, **A. Ghosh**, A. Pramanik, P. Sarkar. "MOFs, COFs and Beyond: A Theoretical Perspective on the Electrocatalytic Activities of 2D Materials toward Sustainable Energy Conversion." Chemical Communications (2026). [Perspective]
4. H. Musah, T. Gamagedara, E. Pathan, A. Afugu, **A. Ghosh**, S.K. Lee, Z.F. Liu, et al. "Generating Long-Lived Highly Excited Carriers in Undoped Perovskite Nanostructures by Two-Photon Upconversion." Nano Letters (2026).
5. A.J. King, F.I.M. Saheed, M. Adeyemo, V. Alevato, D. Kennedy, R. Zhang, et al., incl. **A. Ghosh**. "Resolving Energy Transfer Dynamics at the Quantum Dot Gels–Perylene Diimide Hybrid Interface." The Journal of Physical Chemistry Letters 17 (21), 5990–5997 (2026).
6. M.F. Munoz, D. Konadu, A.M. Sanni, C.L. Ward, **A. Ghosh**, A. Afugu, Z.F. Liu, et al. "Magnetic Response of Excitons and Excitonic Complexes in Defective Hexyl Ammonium Lead Iodide Self-Assembled Quantum Wells." ACS Nano 20 (14), 10794–10806 (2026).
7. **A. Ghosh**, P. Das, D. Maji, D. Barman, P. Sarkar. "Graph Neural Network-Assisted Performance Regulation of Photocatalytic Nitrogen Reduction Reaction: An Insight from Machine Learning-Accelerated Atomistic Dynamics." ACS Catalysis 16 (4), 3284–3295 (2026).
8. S. Kumar, **A. Ghosh**, P. Sarkar. "Unveiling the Photocatalytic Water Splitting over Metal-Free g-CN/γ-BNyne Heterostructures Using Non-Adiabatic Molecular Dynamics." Physical Chemistry Chemical Physics 27 (37), 19724–19732 (2025).
9. S. Kumar, **A. Ghosh**, B. Goswami, P. Sarkar. "Vacancy Defect-Assisted Enhanced Nitrogen Fixation in Metal-Free 2D Orthorhombic Boron Nitride: Insights from Nonadiabatic Molecular Dynamics Simulations." The Journal of Physical Chemistry Letters 16 (32), 8311–8320 (2025).
10. D. Maji, **A. Ghosh**, D. Barman, P. Sarkar. "Accelerating Molecular Dynamics with a Graph Neural Network: A Scalable Approach through E(q)C-GNN." The Journal of Physical Chemistry Letters 16 (9), 2254–2264 (2025).
11. P. Das, **A. Ghosh**, B. Goswami, P. Sarkar. "Theoretical Investigation of a C2N Monolayer as a Bifunctional Electrocatalyst for Rechargeable Non-Aqueous Li–Air Batteries." Journal of Materials Chemistry A 13 (9), 6376–6384 (2025).
12. P. Das, **A. Ghosh**, P. Sarkar. "Photocarrier Dynamics of Two-Dimensional Aza-Fused Covalent Organic Frameworks as Bifunctional Photocatalysts toward Overall Water Splitting." ACS Applied Materials & Interfaces 16 (45), 62043–62051 (2024).
13. S. Kumar, **A. Ghosh**, P. Sarkar. "Promoting Overall Water Splitting in MoS2/BSe Heterostructures: Insights from Time-Domain Atomistic Dynamics." The Journal of Physical Chemistry C 128 (41), 17361–17371 (2024).
14. S. Kumar, **A. Ghosh**, S. Pal, P. Sarkar. "Controlling the Charge Carrier Dynamics of o-B2N2 Monolayer through Pnictogen Family Atoms Doping." The Journal of Physical Chemistry Letters 15 (37), 9388–9396 (2024).
15. **A. Ghosh**, A. Pramanik, S. Pal, P. Sarkar. "Emergence of Z-Scheme Photocatalysis for Total Water Splitting: An Improvised Route to High Efficiency." The Journal of Physical Chemistry Letters 15 (26), 6841–6851 (2024).
16. P. Das, **A. Ghosh**, P. Sarkar. "Hot Carrier Controlled Nitrogen Fixation Reaction in Metal-Free Boron-Anchored Aza-COF: Insight from Nonadiabatic Molecular Dynamics Simulation." The Journal of Physical Chemistry Letters 15 (18), 4898–4905 (2024).
17. **A. Ghosh**, P. Das, S. Kumar, P. Sarkar. "Hot Carrier Relaxation Dynamics of an Aza-Covalent Organic Framework during Photoexcitation: An Insight from Ab Initio Quantum Dynamics." The Journal of Chemical Physics 160 (16) (2024).
18. S. Mandal, **A. Ghosh**, P. Sarkar. "Understanding the Origin of the High Thermoelectric Figure of Merit of Zintl-Phase KCaBi." Physical Chemistry Chemical Physics 26 (17), 13198–13208 (2024).
19. **A. Ghosh**, S. Kumar, P. Sarkar. "Point Defect-Mediated Hot Carrier Relaxation Dynamics of Lead-Free FASnI3 Perovskites." Nanoscale 16 (9), 4737–4744 (2024).

20. **A. Ghosh**, B. Goswami, S. Pal, P. Sarkar. "How the Stacking Pattern Influences the Charge Transfer Dynamics of van der Waals Heterostructures: An Answer from a Time-Domain Ab Initio Study." The Journal of Physical Chemistry Letters 14 (34), 7672–7679 (2023).
21. S. Ghoshal, **A. Ghosh**, P. Roy, B. Ball, A. Pramanik, P. Sarkar. "Recent Progress in Computational Design of Single-Atom/Cluster Catalysts for Electrochemical and Solar-Driven N2 Fixation." ACS Catalysis 12 (24), 15541–15575 (2022). [Review]
22. S. Sannigrahi, **A. Ghosh**, B. Ball, P. Sarkar. "Exploring the Ti2CO2–WSe2 Heterostructure as a Direct Z-Scheme Photocatalyst for Water Splitting: A Non-Adiabatic Study." The Journal of Physical Chemistry C 126 (49), 20852–20863 (2022).
23. **A. Ghosh**, B. Ball, S. Pal, P. Sarkar. "Ultrafast Charge Transfer and Delayed Recombination in Graphitic-CN/WTe2 van der Waals Heterostructure: A Time Domain Ab Initio Study." The Journal of Physical Chemistry Letters 13 (34), 7898–7905 (2022).
24. **A. Ghosh**, S. Mandal, P. Sarkar. "2D Homogeneous Holey Carbon Nitride: An Efficient Anode Material for Li-ion Batteries with Ultrahigh Capacity." ChemPhysChem 23 (15), e202200182 (2022). [Special themed collection]
25. **A. Ghosh**, S. Pal, P. Sarkar. "Rational Design of Two-Dimensional Porous Boron Phosphide as Efficient Cathode Material for Li and Na Ion Batteries: A First-Principles Study." The Journal of Physical Chemistry C 126 (11), 5092–5100 (2022).
26. M. Kar, **A. Ghosh**, R. Sarkar, S. Pal, P. Sarkar. "Arene and Functionalized Arene Based Two Dimensional Organic–Inorganic Hybrid Perovskites for Photovoltaic Applications." Journal of Computational Chemistry 42 (27), 1982–1990 (2021).
27. **A. Ghosh**, M. Kar, C. Majumder, P. Sarkar. "First-Principles Calculations to Investigate Electronic Structure and Transport Properties of CrC Monolayers: A New Horizon for Spintronic Application." Materials Science and Engineering: B 272, 115379 (2021).
28. **A. Ghosh**, M. Kar, C. Majumder, P. Sarkar. "Half Metallicity and Ferromagnetism of Vanadium Nitride Nanoribbons: A First-Principles Study." Physical Chemistry Chemical Physics 23 (2), 1127–1138 (2021).

TECHNICAL & COMPUTATIONAL SKILLS

Electronic Structure & Dynamics Methods: SCF calculations; transition-state search (CI-NEB); optical properties (TD-DFT); ab initio molecular dynamics (AIMD); ballistic transport (NEGF); non-adiabatic molecular dynamics (NAMD) with multiple surface-hopping algorithms

Machine Learning: Graph neural networks (GNN) for accelerated molecular dynamics and property prediction; Python-based ML workflows

Electronic Structure Software: VASP, Quantum ESPRESSO, Gaussian, SIESTA, Hefei-NAMD

Pre-/Post-Processing: VASPKIT, VAMPIRE

Visualization: VESTA, GaussView, Molden, JMol, XCrySDen, XMakemol

Programming & Productivity: Python (intermediate), C, LaTeX, Microsoft Office

Platforms: Linux/HPC clusters, Windows

CONFERENCE PRESENTATIONS

- Participant, "55th Midwest Theoretical Chemistry Conference (MWTCC55)," Wayne State University, Detroit, MI, USA (29–31 May 2025).
- Participant, "Molecular Simulations with RASPA Workshop," University of Massachusetts Amherst (2026).
- Poster Presentation, "Theoretical Chemistry Symposium (TCS-2023)," IIT Madras, Chennai, India (07–10 December 2023).
- Oral Presentation, "Virtual International Seminar on Theoretical Advancements (VISTA)" (05 July 2023).
- Poster Presentation, "National Conference on Molecular Modelling and Simulations (NCMMS 2022)," VIT Vellore, India (28 February – 02 March 2022).
- Poster Presentation, "Emerging Dimensions in Chemical Sciences (EDCS-23)," Kalyani University, India (28–29 March 2023).
- Poster Presentation, "Machine Learning for Science – Symposium and Hands-on Sessions," IIIT Hyderabad, India (29 November – 01 December 2019).

PERSONAL INFORMATION

Date of Birth: 24 November 1995 **Nationality:** Indian **Marital Status:** Married **Gender:** Male

Languages: Bengali (native); English (read, write, speak) and Hindi (read, speak)

Permanent Address (India): Arabinda Pally, Bolpur, Dist. Birbhum, West Bengal 731204, India

REFERENCES

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